

# Arham Mayur Burad

☎ +91 7904727631 | ✉ arhamburad@gmail.com | 🌐 arhamburadd | 📺 roboarhamburad4813

## TECHNICAL SKILLS

---

**Programming Languages:** C, C++, Python, JavaScript

**Web Technologies:** HTML, React.js, Tailwind CSS

**Embedded Systems & Hardware:** Arduino, Microcontrollers, Sensors & Actuators, Circuit Design

**Databases:** MySQL, MongoDB

**Tools & Platforms:** Git, GitHub

## PROJECTS

---

### Thief Detector Robot

- Developed an autonomous surveillance robot using **PIR sensors, and microcontroller-based control (Arduino)**
- Implemented real-time intrusion detection with **event-driven programming** and alert mechanisms (buzzer)

### 5-DOF Robotic Chess Arm with Remote Gameplay | *Ongoing*

- Designed a **5-DOF robotic arm** with **vision-based chess state detection** and real-time actuation
- Developed a custom **PID-controlled PCB** for precise control of **NEMA 17 stepper motors**
- Built a **web-based interface** enabling remote gameplay with synchronized physical move execution

### Colour Sorter Robot

- Developed an automated color sorting robot using **LEGO Mindstorms color sensor** for RGB-based detection
- Designed control logic to classify objects and actuate sorting using **servo motors**
- Optimized sensor calibration and thresholding for accurate real-time object sorting on a **LEGO Mindstorms platform**

### Humanoid Robot

- Designed a humanoid robot on a wheeled base featuring interactive gestures and multi-directional mobility
- Programmed an Arduino in C to manage real-time actuation of **servo and DC motors** using the **Servo.h** library
- Enabled human-like interactions such as handshaking and head movement along moving anywhere we want.

### Smart Dustbin

- Built an automated dustbin system with **touchless operation** using ultrasonic sensing for improved hygiene
- Implemented real-time control on **Arduino Uno (C)** to detect proximity and actuate a servo-driven lid mechanism
- Demonstrated practical application of embedded systems through a working prototype for smart waste management

### HandTrack Assistant

- Built a hand gesture based system using **MediaPipe and OpenCV** for real-time hand tracking and finger classification.
- Developed control logic to map gestures to system-level actions including **brightness, volume, screenshots, and time display**
- Improved interaction reliability by implementing gesture confirmation and cooldown mechanisms

### Olympiad Conduction System

- Built a desktop based exam management system using **Python (Tkinter)** with a structured GUI for registration, evaluation, and result viewing
- Designed and integrated **MySQL database schemas** to manage student data, subjects, and examination records
- Developed automated result computation including grading logic, percentage calculation, and dynamic **state-wise and global ranking**

## EDUCATION

---

### PES UNIVERSITY

2024 – 2028

*Bachelor of Technology in **Artificial intelligence and machine learning***

KA,IN

### MAHARISHI VIDYA MANDIR SSS

2022-2024

*GRADE 11, 12*

TN,IN

### MAHARISHI VIDYA MANDIR SSS

2022

*GRADE 10*

TN,IN

## ACHIEVEMENTS

---

- Secured **1st Runner-Up** in **Line Follower Robot Competition** at Cyber Fest 2019, Maharishi Vidya Mandir, Chennai
- Achieved **Winner** in **Cyber War Event** at Cyber Fest 2020, demonstrating strong problem-solving and technical skills
- Part of **Team SRC Dragsters SRITech Robo Club**, secured **2nd Place** in Drag Race at Roboprix 2K19, VIT Chennai
- Participated in multiple robotics events including **ATM Surprises** and **Speed Trap** at Roboprix 2K19 2.0, VIT Chennai